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Let's Bridge the Gap

SKILL WALL

Coding & Solution

Date 30 September 2026
Team ID SWTID-2026-3226
Project Name FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks 5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission demonstrates working functionality, clean code practices, and alignment with your defined requirements.

- Instructions:
- 1. Ensure the code is well-structured, commented, and follows best practices for the chosen technology stack.
 - 2. The solution addresses the core problem statement and implements the key functional requirements.
 - 3. All source files, configuration files, and setup instructions needed to run the application are included.
 - 4. The code repository is maintained in GitHub for version control and submission.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/kirthip12/FitBuddy-AI-Fitness-Plan-Generator-using-Gemini-Models
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	FastAPI, Uvicorn, Jinja2, SQLAlchemy, Pydantic
Key Features Implemented	AI-powered fitness planning using Gemini Models; personalized 7-day workout plan generation, nutrition and recovery tips, feedback-based plan updates, user profile storage in SQLite, admin view, and demo-mode fallback.
Pending / Incomplete Features	Optional future enhancements: detailed meal plans, progress tracking, user login, mobile-responsive layout, PDF export of plans, multilingual support and advanced customization.
Setup / Run Instructions	Open the “fitbuddy AI” project folder; create a virtual environment; install requirements.txt; add GEMINI_API_KEY to .env; run the FastAPI application with Uvicorn (uvicorn app.main:app --reload).

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

FitBuddy is implemented as a FastAPI web application with modular Python AI components, a SQLite database and a Jinja2 HTML/CSS interface. The project contains source code, configuration, templates, static assets, Gemini integration modules, automated tests and a requirements file. Gemini Models are used to power AI-based workout plan generation, nutrition and recovery tips, and feedback-based plan updates, with a demo-mode fallback when the Gemini API is unavailable.

Implementation Reference:

Repository: github.com/kirthip12/FitBuddy-AI-Fitness-Plan-Generator-using-Gemini-Models | Application: FitBuddy – AI Fitness Plan Generator using Gemini Models | AI Integration: Google Gemini Models | Backend: FastAPI | Database: SQLite | Frontend: HTML and CSS (Jinja2 templates)